

Sihyeon Jegal

Full right to work in the UK (no sponsorship required, Youth Mobility Visa), Leeds, United Kingdom WF17 7SH

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Projects

Portfolio: <https://www.jsihyeon.online>

LeadTaro: <https://www.leadtaro.com>

Github: <https://github.com/wprkftlgus>

Education

2026 University of Ulsan - South Korea

Bachelor of Science BSc, Naval Architecture & Ocean Engineering

[Degree] Graduate

- Completed a comprehensive engineering program focused on advanced mathematics, physics, mechanics, and systems analysis.
- Gained hands-on experience in structural design, CAD modelling, simulation, and computational analysis.
- Built strong expertise in fluid dynamics, thermodynamics, structural analysis, and control systems.
- Improved analytical thinking and data-driven problem-solving through engineering modelling and technical documentation.

Summary

Full Stack Engineer with 5 years of experience building and shipping production web applications. Founded and independently developed LeadTaro, a B2B lead-generation platform featuring real-time data pipelines, improving scraping efficiency from 20 to 25+ results per query and deploying scalable backend systems on AWS with Docker. Experienced in AWS infrastructure migration and deployment workflows, including migrating services and environments to scalable cloud-based architectures with minimal downtime. Recently integrated AI-powered features including cold email generation and intelligent writing enhancement, leveraging modern LLM APIs to automate personalized outreach and improve message quality at scale. Comfortable across the full stack — React/Next.js on the front end, Node.js/Express or .NET on the back end, and AWS/Docker in production.

Certifications

CITA AI Ethics & Regulations Certification (Completed)

Covered key AI ethics principles, including fairness, bias, transparency, and data privacy (GDPR).

Link: <https://docs.google.com/document/d/1V6nkV7pvYbzbuM3WrmicQd5e5GLDqPqd50IESHO5z1c/edit?usp=sharing>

Skills

Languages: JavaScript (ES6+), TypeScript, C#, HTML5, **Cloud & DevOps:** AWS (EC2, ECS, ECR, S3, RDS, CSS3, SCSS, SQL IAM, CloudWatch, Infrastructure Migration), Azure (App Service, SQL Database), Docker (Containerization & Image Builds), Nginx, PM2, GitHub Actions (CI/CD),
Frontend: React.js (Hooks, React Router, SPA), Next.js, Angular, Vite, Tailwind CSS, Bootstrap, Three.js Certbot (HTTPS/SSL), Hetzner Cloud (Worker Service Deployment), Redis-based Task Queues
Backend: Node.js, Express.js, ASP.NET Core Web API, REST APIs, Middleware, CORS **Auth & Security:** JWT, Cookies, Bcrypt, OAuth 2.0 (Google Authentication), Environment Configuration (.env)
Databases: MongoDB (Mongoose), MySQL, MsSQL, Entity Framework Core, AWS RDS, Upstash Redis **Tools:** Git, GitHub, Visual Studio Code, Microsoft Office, Netlify, Vercel
AI Integration: Experience integrating LLM APIs for automated content generation (e.g., cold email generation with prompt engineering and retry handling)

Practices: Responsive Web Design, Mobile-First Development, UX/UI Fundamentals, Distributed System Awareness, Unit/Integration Testing Awareness

Experience

Packaging Operative – PPG (Amazon Packaging), 06/2026 - Current

PPG - Batley, United Kingdom

- Assembled shipping boxes and packaging materials for paint products.
- Packed, sealed, and prepared products for dispatch.
- Stacked and palletized packaged goods safely and efficiently.
- Performed quality checks to ensure packaging accuracy and product protection.
- Followed health, safety, and warehouse procedures in a fast-paced environment.

LeadTaro, AI Driven Full stack Developer, 10/2025 - Current

LeadTaro - batley, United Kingdom

- Built and launched a B2B lead-generation SaaS platform from scratch, covering product design, **full stack** development, cloud deployment, and growth using **TypeScript, Next.js, React (Hooks), Tailwind CSS, HTML, Node.js, Express.js, and MySQL**.
- Integrated an AI-powered cold email generation system using **LLM APIs**, enabling automated and personalised outreach with custom prompt engineering and retry handling for reliability.
- Added an AI-assisted writing enhancement feature to refine grammar, tone, and clarity of generated emails, improving overall message quality and user experience.
- Leveraged **AI-driven** solutions to accelerate development while ensuring secure, maintainable, and production-ready code.
- Managed and **migrated AWS** infrastructure and services across environments, including container deployments, configuration updates, and production infrastructure improvements with minimal downtime.
- Architected a distributed system separating an API server (**AWS ECS + Docker**) from a background worker service (Hetzner) coordinated via Upstash Redis queues for heavy scraping tasks. Stored environment variables securely in **AWS S3** directed infrastructure entirely as sole developer.
- Built a real-time data extraction pipeline for location-based search, retrieving **20–25+** business records per query and improving result yield by **25%** through scroll-timing optimisation.
- Planned and executed SEO strategy from scratch, achieving **20%+** organic CTR and steadily increasing search impressions within the first month.
- Integrated Stripe payments and **Google OAuth**; secured auth with **JWT, bcrypt, and cookie** sessions.
- **ECR** image builds, **EC2 load-balancer** health checks, **CloudWatch** monitoring, and internal **SSL** certificates.

Software Engineer(Frontend), 05/2023 - 12/2025

M Star Factory - Seoul, South Korea

- Developed a dermatology clinic website from scratch using HTML, CSS, JavaScript, and React
- Built responsive and reusable UI components, improving mobile usability across devices
- Optimized performance and page load speed, enhancing overall user experience
- Conducted UX testing and applied improvements based on user feedback
- Collaborated with a developer to deliver the project end-to-end

Co Owner & Property Manager, 03/2024 - 09/2024

Mithalia ltd - Leeds, United Kingdom

- Operated and supervised a short-term rental property end-to-end, including guest communication, listings, and pricing
- Communicated with domestic and international guests to handle enquiries, bookings, and issue resolution
- Produced high-quality property photos and optimized listings to improve visibility and booking rates

- Promoted listings across multiple platforms to maximize occupancy
- Managed calendars, availability, and day-to-day operations independently

Military Security Specialist & Transport Operator, 11/2021 - 05/2023

South Korean Military - Siheung, Gyeonggi Province, South Korea

- Demonstrated discipline, reliability, and teamwork in a high-responsibility environment.
- Conducted security patrols, access control, and operational reporting.
- Operated and maintained transport vehicles for personnel, and equipment.
- Warned violators of rule infractions and escorted unauthorised people off-premises in line with company procedures.

Languages

English:

C1

Advanced

Korean:

C2

Proficient